

Financial Markets: 19. Investment Banks

Robert J. Shiller

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Original lecture by Robert J. Shiller. Transcript-derived notes curated by [LazyingArt LLC](#).

Chapter 1

Investment Banks

These notes follow Robert J. Shiller's lecture on investment banking, together with Jan Fugner's classroom discussion, and have been curated by LazyingArt LLC. The lecture begins not with a formula but with a distinction. We first ask what investment banking is, and we answer by separating it from consulting, trading, and commercial banking. Only then does Shiller narrow the picture to underwriting, due diligence, and reputation. From there the lecture moves outward to regulation and inward again to the crisis mechanism of repo funding and shadow banking. After that first half, the lecture deliberately changes scale: Jan Fugner takes us inside the analyst's work, then out again to Facebook, the social graph, click-through rates, and the broader claim that finance and engineering share a discipline of rigorous design.

1.1 The Business Called Investment Banking

Let us begin where the lecture begins. Investment banking is a business of helping other entities create securities. A corporation may want to issue stock; a government may want to issue bonds; a nonprofit or some other incorporated entity may also need access to capital markets. In each case the investment bank stands between the issuer and the outside investor and helps turn a financing need into a marketable security.

Definition 1.1. An investment bank, in the narrow sense used at the start of the lecture, is an institution that helps issuers create and place securities. A pure investment bank does not accept deposits.

That last point matters, because Shiller frames the whole topic by contrast. The investment bank shares something with consulting, since it may advise on corporate strategy and financing choices. But it is not a pure consulting firm, because its advice comes attached to capital-market execution. It shares something with banking, since it helps solve financing problems. But it is not, in its pure form, a deposit-taking institution. And it differs from trading because the trader buys and sells for market purposes, whereas the investment banker is chiefly concerned with helping the client create or place a security.

The lecture then makes the definition operational by moving immediately to underwriting. A company that wants to issue shares needs someone who knows the investor base, can organize distribution, and can in some degree vouch for the offering. That is the underwriting function.

Definition 1.2. Underwriting is the process by which an investment bank helps issue and place securities, sometimes standing behind the sale more strongly and sometimes less strongly, depending on the form of the deal.

From there the lecture names the standard objects:

- An initial public offering is the first public sale of a firm's shares.
- A seasoned offering is a further issue by a firm whose shares already trade publicly.
- In a bought deal, the bank buys the securities from the issuer and then resells them.
- In a best-efforts offering, the bank does not buy the securities itself but undertakes to place them as successfully as it can.

Shiller also notes that the mechanics of issuance are regulated, in the United States by the Securities and Exchange Commission. So even at this early stage the lecture is preparing us for an institutional story, not merely a dictionary definition.

1.1.1 Question & Answer

Question. Why is investment banking not just consulting with better contacts, or just another kind of banking?

Answer. Because the function is different. A consultant may give advice but does not necessarily bring capital. A commercial bank may take deposits and make loans but does not primarily organize securities issuance. The investment bank combines advice with market access: it helps define the problem, structures the financing, and then brings the issuer into contact with investors through securities markets.

1.2 Underwriting as a Trust Technology

Now the lecture asks the next question, and it is the right one. Why is such an intermediary needed at all? Why can the issuer not simply go straight to the public?

Shiller's answer is moral hazard. The issuing firm knows more about itself than outside investors do. If things are about to go badly, insiders may have the strongest possible incentive to issue claims before the market fully understands the situation.

Definition 1.3. In this setting, moral hazard means that insiders may exploit their superior information by selling securities before adverse information becomes public.

The lecture puts the point almost brutally: if the company knows it is close to failure, why not issue shares now and "milk" the firm before the public finds out? Underwriting matters because the bank is supposed to block exactly this sort of opportunism. It performs due diligence, checks the issuer, and places its own reputation at stake.

So we arrive at the deeper claim of the lecture: investment banking is built around trust. Underwriting is not merely a distribution service. It is a trust technology. The bank lends its reputation to the

issuer and, by doing so, lowers the fear that the offering is simply a transfer from better-informed insiders to less-informed outsiders.

This is why Shiller lingers on manners, presentation, and cultivated seriousness. He is not merely offering social comedy. He is saying that, in this business, reputation is an operating asset. Investors, clients, and senior executives have to believe that the intermediary is selective, informed, and careful.

The Goldman Sachs discussion then enters exactly at this point. Shiller uses Charles Ellis's *The Partnership* not to wander off into corporate anecdote, but to give the trust argument an institutional body. Some of the Whitehead principles strike him as almost platitudinous when read in isolation. Yet at Goldman they seem to have worked as discipline. In the lecture, the most important themes are these:

- clients' interests come first;
- the firm's real assets are people, capital, and reputation;
- loyalty and discretion matter;
- one should deal with actual decision-makers, not merely their assistants;
- publicity is often shunned rather than sought.

The lecture is careful here in its own way. Shiller does not simply praise Goldman. He notices the severity of the culture and the possibility that some listeners will be repelled by it. But he keeps returning to the same economic point: a bank whose business rests on reputation must cultivate habits that protect that reputation.

1.2.1 Question & Answer

Question. Why does underwriting depend so heavily on trust and institutional reputation?

Answer. Because the bank's value lies precisely in its ability to stand between issuer and investor and say, in effect, that the issue has been examined. If the bank's name means nothing, then its presence adds little. If its name is valuable, then due diligence and careful client selection become part of the product itself.

1.3 Glass-Steagall, Universal Banking, and Partial Separation

Once the lecture has shown us what investment banks do, it turns naturally to the state's problem. If some institutions take insured deposits and others underwrite securities or trade on their own account, where should the line be drawn?

Shiller answers first with history. In 1933 the Glass-Steagall framework separated commercial banking from investment banking. This was linked to the creation of federal deposit insurance through the FDIC. The logic is straightforward. If the state insures deposits, then it acquires an interest in constraining the risks taken by insured institutions. Deposit insurance without activity restrictions would be an invitation to moral hazard at the level of the bank itself.

So the old separation had a practical point: insured deposit banking was not to be mixed freely with activities regarded as more speculative or dangerous. Shiller then uses the Morgan example to

illustrate the forced separation. The historical genealogy is spoken loosely in the lecture, and we need not make it more precise than the lecture does. The important point is that institutions had to choose which side of the line they would inhabit.

The lecture then jumps ahead to repeal. In 1999, under what Shiller refers to loosely as the Graham-Leach legislation, the United States moved back toward universal banking. He emphasizes the competitive argument. Outside the United States, many countries had long permitted banks to conduct both commercial and investment banking. American banks increasingly looked constrained by comparison, and repeal was defended in part as a response to that disadvantage.

But then came the crisis, and the old question returned. If insured or quasi-insured institutions can engage in increasingly risky market activities, should the line be redrawn?

Shiller's answer is not that the country simply restored Glass-Steagall whole cloth. Instead he describes partial barriers. The Volcker Rule is the central case. Commercial banks are restricted from proprietary trading and from certain relationships with hedge funds and private-equity vehicles. He also points to the Lincoln amendment as another attempt to keep some swap-related activity away from the federal safety net. The lecture's statutory details are loose, but the policy thrust is clear: rather than separating institutions completely, post-crisis regulation tried to separate protected funding from especially dangerous activities.

One of Shiller's broader remarks belongs here. Finance, he says, is a profoundly rules-based world. The big bills get the headlines, but the real content lives in thick legal detail. That observation explains some of the lecture's tone in this section: the issue is not simply whether one favors regulation, but where the practical boundary can be drawn in an enormously elaborate legal environment.

1.3.1 Question & Answer

Question. Why did the state separate commercial and investment banking in the first place, and why only partially restore that line later?

Answer. The original separation followed from deposit insurance. Once deposits are publicly guaranteed, the state has to care what insured institutions do. After repeal, however, banking had already become more integrated and more globally competitive. Post-crisis reform therefore aimed less at total divorce and more at blocking certain combinations, especially insured funding plus proprietary speculation.

1.4 Shadow Banking, Repo Funding, and the Run on Lehman Brothers

Now the lecture reaches its sharpest analytical point. We are no longer asking only what the law calls a bank. We are asking what the institution does economically. A firm may not take deposits and yet may still behave like a bank in the most important sense: it may fund long or risky assets with short-term runnable liabilities.

Definition 1.4. A shadow bank is an institution that performs economically bank-like funding functions without being regulated as a commercial bank.

Shiller then introduces Lehman Brothers as the case that makes this idea concrete. Lehman was, in the legal sense, a pure investment bank. It did not take deposits and was not regulated like a commercial bank. Yet Gary Gorton's interpretation, which Shiller cites, is that Lehman had come to fund itself in a bank-like way through the repo market.

Definition 1.5. A repo, or repurchase agreement, is a transaction in which a security is sold today with an agreement to repurchase it later. Economically, it functions as a short-term collateralized loan.

The lecture states this verbally rather than in notation, so let us introduce a cleaned formalization carefully.

Remark 1.6. The following notation is a standard reconstruction of the funding logic described in the lecture. It is not presented as a visible board derivation.

$$\text{sale today : collateral} \mapsto P_0, \quad (1.1)$$

$$\text{repurchase later : } P_1 \mapsto \text{same collateral}. \quad (1.2)$$

From this we obtain the simplest pricing summary:

$$\text{financing cost} = P_1 - P_0, \quad (1.3)$$

$$r_{\text{repo}} = \frac{P_1 - P_0}{P_0}. \quad (1.4)$$

Shiller's larger point is not the rate formula itself, but the liability structure. Repo borrowing is short term. It can disappear quickly. In that sense, as he says explicitly, it is "almost the same as a deposit":

$$\text{repo funding} \approx \text{short-term runnable deposit-like funding}. \quad (1.5)$$

The lender is different. It is not a household with a savings account but typically an institutional counterparty. Still, the economic vulnerability is similar. If the lender fears failure, it does not renew the funding. In the commercial-bank case the depositor withdraws. In the repo case the lender refuses rollover.

This gives us the compact liquidity identity behind the lecture's run mechanism:

$$\text{maturing repo not renewed} = \text{asset sales} + \text{new funding}. \quad (1.6)$$

If new funding cannot be found, assets must be sold. If the assets are already under pressure, forced sales worsen the situation. The institution now looks very much like a bank facing a run.

1.4.1 Question & Answer

Question. How can a firm that does not accept deposits still collapse in something that looks exactly like a bank run?

Answer. Because the essence of the run is not the legal form of the liability but its maturity and renewability. If a firm's funding is short term and confidence-sensitive, then fear can remove that funding all at once. Deposits are one example. Repo borrowing is another.

The lecture's causal chain can be written as a short derivation.

Derivation of the Lehman mechanism.

1. Lehman finances a large quantity of assets through short-term repos.
2. Those assets include subprime-linked and other risky securities.
3. The housing decline lowers the value of those assets and of the collateral behind the funding.
4. Repo counterparties lose confidence in Lehman's condition.
5. They stop renewing maturing repos.
6. Lehman must replace the funding immediately or sell assets into a weak market.
7. Without sufficient rescue, the failure of rollover becomes the failure of the firm.

That is why Shiller calls the 2008 crisis, in substantial part, a "run on the repo." The legal category of the institution mattered less than its economic resemblance to a bank. The post-crisis lesson, as he states it, is that shadow banking cannot simply be left outside the regulatory picture.

1.5 Jan Fugner: Analyst Work and the Debt-Boom Tempo

Only after laying out that institutional and crisis logic does Shiller bring Jan Fugner to the front. The handoff matters. First we are shown what investment banking is supposed to do in the system. Then we hear what it was like to work inside one of those institutions.

Jan begins with a bit of personal background, but he very quickly narrows to the technical core of junior banking. If one wants a one-word emblem of the analyst's life, his answer is essentially Excel. The junior banker is not, at first, a grand strategist. He is the person who builds the numerical machinery on which the senior relationship depends.

The lecture names three model types again and again:

- operating models,
- transaction models,
- valuation models.

These are then turned into polished pitch books, used to win business, and, once the business is won, used to help move the deal across the finish line. Jan's description is valuable because it preserves the executorial character of the work. The analyst deals with accountants, lawyers, other bankers, clients, and counterparties. In a lean team the analyst is often the organizing center through which the details pass.

The team structure is presented in the lecture in the simplest ordered form:

$$\text{Analyst} \rightarrow \text{Associate} \rightarrow \text{VP} \rightarrow \text{MD}. \quad (1.7)$$

That is not an algebraic law, but it captures the practical layering of responsibility. The core deal team may have only one person at each level. This is why Jan emphasizes both pressure and opportunity: a junior banker can take on unusually large responsibility if the work is done accurately.

1.5.1 Question & Answer

Question. Why is junior banking presented as a relationship business but experienced as a modeling business?

Answer. Because the relationship wins access, but the model earns credibility inside the work itself. Senior bankers may cultivate CEOs and CFOs, but the analyst must still build the operating, transaction, and valuation models, gather the comparables, prepare the pitch book, and make sure that the numbers survive contact with the lawyers, accountants, and counterparties.

Jan also preserves the tempo of the pre-crisis boom. Private-equity recruiting had become feverish. Analysts were being approached and sometimes committed to future jobs far earlier than the formal timeline would suggest. Transactions were so frequent and so large that people spoke of “Merger Mondays” as though a new multibillion-dollar announcement at the start of the week had become part of the calendar.

The exuberance went further. Even financial institutions, already heavily levered, began to look like possible leveraged-buyout targets. Jan marks Blackstone’s public offering as a kind of symbolic peak of the era: one of the great buyout firms itself becoming a public company.

The discussion of deal toys is lighter, but it still serves a purpose if we keep it in proportion. It reminds us that analyst life mixed precision, exhaustion, ritual, and a certain forced creativity. We should not let that joke dominate the chapter, but neither should we erase the texture. It belongs to the inside description of the job.

Then the chronology darkens very quickly. Bear Stearns is sold under pressure. Merrill Lynch avoids liquidation by folding into Bank of America. Lehman collapses. Goldman Sachs and Morgan Stanley convert to commercial-bank status. Jan’s witness here is useful because it gives the same break in tone that Shiller had just analyzed from the outside: the old Wall Street structure really did come to an end.

1.6 From Wall Street to Facebook: Graphs, Funnels, and Rigor

Jan’s move to Facebook is not a departure from the lecture’s main theme. It is the lecture’s last reframing of that theme. What changes is the object of analysis, not the demand for disciplined reasoning.

One of the clearest late moments in the lecture comes when Shiller asks Jan how Facebook’s core values compare with Goldman’s. Jan’s answer is precise. Each institution has a core constituency. At Goldman the focal point was the client. At Facebook the focal point is the user and the user experience. That distinction matters, because it explains why the second half of the lecture moves from securities and balance sheets to graphs, advertising, and product design.

The social graph is Jan’s mathematical shorthand for the platform’s structure. Again, the lecture

states the idea verbally, so we formalize it cautiously.

Remark 1.7. The graph notation here is a cleaned reconstruction of Jan Fugner’s description of Facebook’s task as drawing a mathematical representation of “who likes whom” and “who likes what.”

$$G = (V, E), \quad (1.8)$$

or, if we wish to distinguish people from objects,

$$G = (U \cup O, E). \quad (1.9)$$

The point is not graph theory for its own sake. The point is that information can be targeted and transmitted along relationships already present in the graph. That is why social context can matter in advertising: it changes not only who sees a message, but how the message is interpreted.

Jan then turns to the most explicit numerical definition in this half of the lecture, the click-through rate:

$$\text{CTR} = \frac{\text{clicks}}{\text{impressions}}. \quad (1.10)$$

He immediately gives the lecture’s worked example:

$$\frac{1}{100} = 1\%. \quad (1.11)$$

So if an advertisement is shown 100 times and receives one click, its click-through rate is 1%. But Jan is careful not to let the class confuse a clean ratio with a complete measure of advertising value.

The right framework, he says, is the marketing funnel:

$$\text{awareness} \rightarrow \text{affinity} \rightarrow \text{consideration} \rightarrow \text{purchase} \rightarrow \text{repeat purchase}. \quad (1.12)$$

At the bottom of the funnel, where the consumer is already searching for a product and close to buying, CTR may be highly informative. A click is then a natural sign of movement toward purchase. But higher in the funnel the advertiser may be trying to generate awareness or affinity instead. In that case one may need different measurements, including survey-based or polling-based measures of brand lift.

Jan also gives two useful scale markers:

$$\text{reach} \approx 500 \text{ million users}, \quad (1.13)$$

$$\text{online advertising share} \approx 15\%. \quad (1.14)$$

Those figures do not constitute a theory, but they explain the strategic claim: Facebook advertising can operate at many points in the funnel because it combines broad reach, targeting, and social context.

1.6.1 Question & Answer

Question. Why is click-through rate only one metric, and why does “thinking like an engineer” matter even outside engineering?

Answer. CTR measures one kind of response, and only one. If the relevant task is bottom-of-funnel conversion, it can be very informative. If the task is awareness, affinity, or message recognition, other measures may matter more. The engineering analogy enters because the real discipline is to match measure to mechanism. One must define the objective, understand what the system can actually do, and then use data rigorously enough to test whether the intervention is working.

That last point connects directly with Jan’s broader reflection about working in technology. One need not be a software engineer to contribute in an engineering company. But one does need to think with engineering-style rigor: what is feasible, what is measurable, what experiment would distinguish one claim from another, and what evidence is strong enough to change a decision. Shiller embraces that connection explicitly in the final minutes, remarking that finance and engineering both design devices.

1.7 Summary

The lecture unfolds in a deliberate sequence. We begin with a narrow institutional definition: the investment bank helps issuers create and place securities, and underwriting is its central mechanism. We then see why that mechanism exists at all: it answers a trust problem by combining due diligence with institutional reputation. From there the lecture asks where the regulatory boundary should lie once deposit insurance, universal banking, and speculative activity begin to mix. The answer is only partly legal, because shadow banking shows that an institution can avoid the name “bank” while still reproducing the funding fragility of a bank.

That is why the repo discussion sits at the center of the chapter. A firm financed by short-term runnable repo liabilities can suffer the economic equivalent of a bank run even if it takes no deposits. Jan Fugner’s testimony then changes the scale without changing the intellectual style. Junior banking is a world of models, pitch books, lean teams, and boom-time deal pressure. The move to Facebook then generalizes the lecture’s deeper lesson: whether we are underwriting a security, rolling repo funding, mapping a social graph, or measuring advertising, the work becomes serious only when we make the mechanism explicit and reason through it carefully.